Curso de Tecnologia em Sistemas de Computação
Matemática para Computação - EAD05008
Lista de exercícios sobre funções I - Gabarito

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1. (a) $A = \{x \in \mathbb{R} \mid x > 0\}$
   (b) $B = \{x \in \mathbb{R} \mid 5 \leq x < 3\} = \emptyset$
   (c) $C = \{x \in \mathbb{R} \mid -1000 \leq x < 100 \ \vee \ 200 < x < 300\}$
   (d) $D = \{x \in \mathbb{R} \mid -5 < x \leq -1\}$
   (e) $E = \{x \in \mathbb{R} \mid -6 \leq x \leq 5 \ \vee \ x = -1 \ \vee \ x = 2\}$

2. (a) $A = (-3, 7)$
   (b) $B = [0, 0] \ \cup \ [5, 5]$
   (c) $C = [0, 5]$
   (d) $D = [-1, 20]$
   (e) $E = (-20, 3] \ \cup \ [4, 9]$

3. (a) $Dom(f) = \mathbb{R}$ e $Im(f) = [-4, +\infty)$
   (b) $Dom(f) = [-1, 1]$ e $Im(f) = [0, 1]$
   (c) $Dom(f) = [6, +\infty)$ e $Im(f) = [17, +\infty)$
   (d) $Dom(f) = \mathbb{R}\backslash\{0\}$ e $Im(f) = (-\infty \ , \ -6,828427\cdots] \cup [-1,171572\cdots \ , \ +\infty)$

4. (a) $(f \circ g)(x) \ = \ (3x-2)^3 \ = \ 27x^3 - 54x^2 + 36x - 8$
   (b) $(u \circ g)(x) \ = \ \dfrac{1}{(3x-2)-2} \ = \ \dfrac{1}{3x-4}$
   (c) $(u \circ u)(x) \ = \ \dfrac{1}{\left(\dfrac{1}{x-2}\right) - 2} \ = \ \dfrac{1}{\dfrac{1-2(x-2)}{x-2}} \ = \ 1 \cdot \dfrac{x-2}{1-2(x-2)} \ = \ \dfrac{x-2}{5-2x}$
   (d) $(f \circ g \circ u)(x) \ = \ \left[3\left(\dfrac{1}{x-2}\right) - 2\right]^3 \ = \ \left(\dfrac{3}{x-2} - 2\right)^3 \ = \ \left(\dfrac{3-2(x-2)}{x-2}\right)^3 \ = \ \left(\dfrac{7-2x}{x-2}\right)^3$
   (e) $(h \circ f)(x) \ = \ \text{sen}(x^3)$
   (f) $(h \circ g)(x) \ = \ \text{sen}(3x-2)$